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Part 1) Timing Analysis (20%)

1.1 - Determine Machine Architecture

EXERCISE 1: 12 and 28.

Discussion Question 1

	Cache Line Size	Total Size	Number of Ways (Associativity)	Number of Sets	Raw Latency
L1	64 Bytes	32,768 Bytes	8	64	4
L2	64 Bytes	1,048,576 Bytes	16	1024	12
L3	64 Bytes	11,534,336 Bytes	11	16,384	40

Table 1: Cache Parameters

1.4 - Timing a Memory Access

1) Measure L1 Latency

EXERCISE 2:

L1 Latency Measurements:

1. Minimum: 110
2. Bottom decile: 110
3. Median: 114
4. Top decile: 114
5. Maximum: 114

2) Measure DRAM Latency

The SSH environment wouldn't let us run the reference file, even after running chmod like we had to for victim-N.

```
arastrel@burnsville:~/snap/snapd-desktop-integration/current/Desktop/590-WebsiteFingerprinting/hwsec-course/lab-cacheattacks/Part1-Timing$ make run-reference
taskset: failed to execute ./reference: Permission denied
make: *** [Makefile:9: run-reference] Error 126
```

EXERCISE 3:

DRAM Latency Measurements:

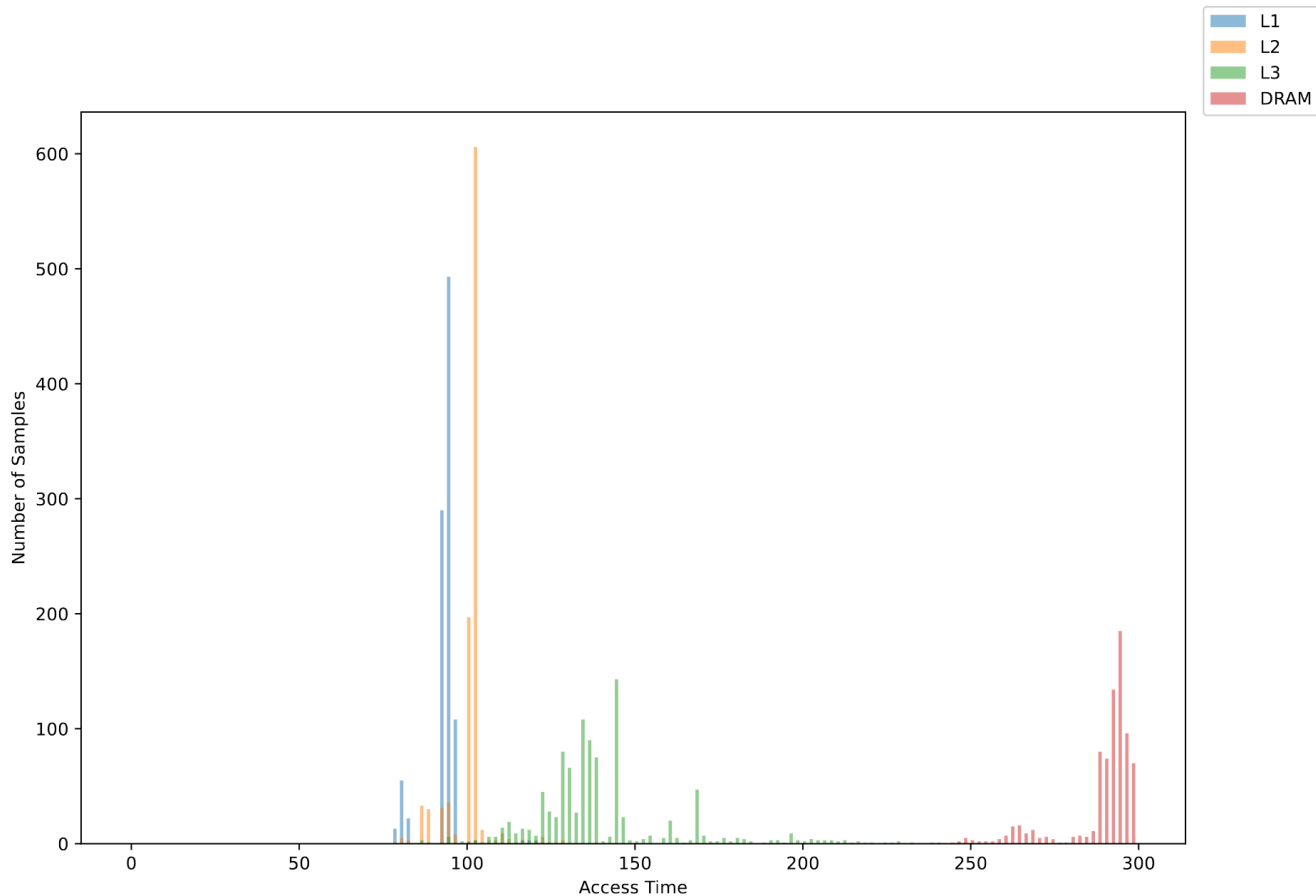
1. Minimum: 290
2. Bottom decile: 322
3. Median: 330
4. Top decile: 346
5. Maximum: 346

3) Measure L2 and L3 Latency

- a) **Discussion Question 2:** How many cache lines do you need to access to fill up the entire L2 cache and L3 cache respectively? You can derive the answer from Table 1. Put your answer in 1.4.txt
 - i) To fill up the entire L2 cache and L3 cache, you would need 16,384 cache lines and 180,224 cache lines respectively.
- b) **EXERCISE 4**
 - i) L2 Median: 122
 - ii) L3 Median: 156

1.5 - Visualize Latency Distribution

- a) **EXERCISE 5 (next page)**



b) Discussion Question 3: Based on the generated histogram, report two thresholds, one to distinguish between L2 and L3 latency and the other to distinguish between L3 and DRAM latency. Put your answers in 1.5.txt.

A threshold to distinguish between L2 and L3 latencies would be around 110, while a threshold to distinguish between L3 and DRAM would be around 230.

Part 2) Dead Drop: An Evil Chat Client (50%)

2.2 - Transmitting a Single Bit Across Processes

DONE

2.3 - Transmitting An 8-bit Integer Across Processes

Discussion Question 6: Describe your communication protocol.

- 1) Encode: The sender's action for sending bit 1 is to ensure minimal latency in the cache line that the sender and receiver are both watching. We did this by using an mmap and repeatedly accessing a set memory location. For bit 0, we evicted all of the data in the L1 and L2 caches to guarantee high latency.
- 2) Receive: The receiver is looking at the same mmap location that the sender is repeatedly hitting. Depending on the latency produced by reading what is at that mmap location, the receiver knows which bit the sender is trying to convey.
- 3) Threshold: The threshold we came up with for differentiating between a 0 and a 1 was a latency of 130. We found this to work best to tell if an access came from L1/L2 cache or if it came from L3 cache.

EXERCISE 7: Implemented!

3 - Capture The Flag

EXERCISE 8: Implemented!